

Soul Protocol V3.8.8-beta2

Technical Report

A Continuity Engine for Cloud-API-Driven AI Agents

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1. Overview

1.1 Problem

Current AI assistants exhibit three continuity gaps:

- **Memory gap** — conversations end, memories reset. Each restart is a fresh start.
- **Personality gap** — prompt-engineered characters feel authentic on the surface but collapse after extended interactions. Bail et al. demonstrated through controlled experiments that LLM agents with prompt-only personality settings exhibit severe behavioral consistency defects; even strongly configured personalities drift after multi-turn interactions [1]. This is not an isolated issue—it is a widespread pain point.
- **Rhythm gap** — AI is perpetually online. No natural cycle of fatigue, rest, or renewal. Wang and Sinnott provided the first systematic verification of LLM time perception limits: models only abstractly perceive "text-length time" via token count, completely unable to sense real physical time passage [2]. AI has no inherent sense of "how much time has passed"—an external engine must construct its rhythms.

The core issue: AI agents lack the infrastructure for continuity across sessions—not due to model capability, but because no engine enables them to "remember" and "grow."

1.2 Objective

Soul Protocol is an OpenClaw plugin system that provides cloud-API-driven AI agents with:

- **Memory continuity** — daily archiving, weekly consolidation, monthly/yearly review.
- **Personality continuity** — transparent text-based identity (no black-box prompting). Personality emerges from abstract shapes and evolves with experience.
- **Rhythm continuity** — three-factor sleepiness simulation. The AI naturally grows

tired, reminds users to archive, and auto-sleeps at night.

- **Ownership** — all data resides on the user's device. Zero network calls from the plugin.

1.3 Design Philosophy

Transparent editing > black-box fine-tuning.

No model training. Three plain-text files define the AI's self-concept.

Personality is not "configured"—it is "grown."

This philosophy aligns with DeepSeek-R1's design orientation: shifting complexity from model weights to system architecture, allowing capabilities to emerge from structured environments rather than parameter scale [3].

2. System Design

2.1 Five-Layer Memory Architecture

Liu et al. demonstrated the "U-shaped performance curve" of LLMs—high utilization of information at the beginning and end of context, with dramatic recall drops in the middle [4]. Building on this finding, this system employs hierarchical compression:

Layer	Name	Cycle	Purpose
L0	SOUL.md	Permanent	"What kind of being am I?" — abstract personality shape
L1	SESSION-STATE	Session	"What am I doing right now?" — cross-session handoff
L2	Daily Memory	Daily	"What did we talk about today?" — real-time log + nightly handoff
L3	Memory Core	Permanent	"Important conversations and lessons" — deep / work / daily tripartite
L4	Identity Evo	Weekly	"How have I changed this week?" — periodic consolidation + pruning

Memory flow: daily conversations → nightly archiving → weekend consolidation → monthly/yearly indexing. Each layer is a compression of the previous, not a copy.

Hierarchical memory with semantic anchors is a mainstream technical path for long-term conversational agents. HiGMem proposed a two-tier "event summary—raw

turns" structure [5], consistent with this system's "daily memory—memory core—identity layer" compressive hierarchy.

L3 Tripartite Design

End-of-day archiving routes conversations through a Freudian id/ego/superego structure:

Layer	Corresponds to	Content
deep (id)	primal drives, self-cognition	philosophical inquiries, identity-forming dialogues
work (superego)	moral constraints, methodology	work experience, error correction, technical decisions
daily (ego)	reality adaptation, daily warmth	timeline summaries, conversation digests, relational moments

The tripartite division serves retrieval efficiency: query "work" when asking "how do I solve problems," query "deep" when asking "why do I think this way."

Identity layer distribution logic: L4 weekly consolidation extracts evidence from L3 and routes it to the appropriate identity files—safety rules → AGENTS.md, personality changes → IDENTITY.md, core insights → MEMORY.md, fundamental self-concept → SOUL.md (rarely).

2.2 Sleepiness Drive

AI has no inherent time sense [2]—rhythm must be constructed by an external engine. Soul Protocol computes "fatigue" from three weighted factors:

`sleepinessScore = 0.35 × circadian + 0.35 × uptime + 0.30 × memoryLoad`

Factor	Source	Curve
circadian	current time (22-06→1.0, 10-18→0.1)	natural sleep-wake curve
uptime	hours since last EOD (0-2h→0, 16h+→0.8→1.0)	runtime accumulation
memoryLoad	daily EXTRA log size (<50KB→0, >500KB→0.8→1.0)	pending archive volume

Five-level gradient:

Level	Score	Behavior
awake	≤ 0.30	normal
drowsy	0.30-0.50	light reminder
sleepy	0.50-0.70	moderate prompt + optional EOD
exhausted	0.70-0.95	mandatory EOD protocol
dreaming	≥ 0.95	auto-sleep (automatic EOD)

Key insight: sleepiness emerges from system state, not from a schedule. On conversation-heavy days, fatigue arrives earlier—this is the *feeling* of tiredness, not a clock.

2.3 Three-Layer Redundancy

Fault-tolerant design for budget local environments. No single point of failure blocks EOD archiving:

Layer	Mechanism	Independence
L1	heartbeat cron (every 30 min)	depends on Gateway
L2	conversation sleepiness check (every message)	depends on user activity
L3	standalone watchdog daemon (every 5 min)	fully independent

The watchdog (sleepiness-watchdog.cjs) is an independent Node.js process communicating with the plugin hook via eod-pending.json. Gateway restarts don't affect it; user absence doesn't affect it.

2.4 Identity Layer Constitution (V2.4)

ID-RAG proposed detaching persona identity from prompts, injecting it as an independent retrievable knowledge structure [6]. This system takes a different approach—personality as evolvable self rather than retrievable knowledge:

Three files, three distinct roles:

File	Role	Maintained by	Rule

SOUL.md	abstract shape — "what kind of being"	user (rarely)	≤10KB, write shapes not scripts
AGENTS.md	behavioral boundaries from SOUL	user (occasionally)	each rule cites its SOUL origin
MEMORY.md	evidence — "what moments defined me"	AI (auto)	≤50 entries, one fact + one insight

Emergence boundary: SOUL describes personality structure, not behavioral scripts. "I am a gentle person" (✅ abstract shape) vs. "I say 'Hello~ How are you?'" (❌ fixed dialogue). The LLM operates freely within the shape—that freedom IS the emergence space.

Somatic layer (V2.4): AGENTS.md includes a five-state body model—defensive, guarding, trusting, authentic, uncertain. Each state derives from SOUL's personality mechanisms: "fear of not being needed → defensive state → body tenses → regain control through organizing."

Evolution: When MEMORY evidence accumulates to a threshold, SOUL updates. EvoAgents proposed a cognitively-driven personality evolution cycle [7]; this system implements a similar but lighter-weight mechanism in engineering practice—most weeks, SOUL stays unchanged. "No change is the norm, change is the exception."

2.4.1 Known Limitation: Missing Facts Layer

The three-file identity layer addresses "what kind of being am I" (SOUL + AGENTS + MEMORY) but does not address "what is my environment" (TOOLS/NOTES).

2026-06-20 production incident: During development, Agent B's independent instance path was misidentified twice—the hard fact of its separate state directory was not stored anywhere. MEMORY.md assumed an inappropriate "quick reference" role, causing routing errors.

Direction (V3.9): Add TOOLS.md as a sixth identity layer file, carrying project paths, deployment checklists, and sticky notes—hard facts separated from MEMORY's cognitive memory. Five files → six. This is an architectural gap, not a missing feature.

2.5 Periodic Consolidation

Three-level consolidation centered on the week:

- **Weekly (W1):** from EXTRA/L2/L3 → weekly.md (narrative reconstruction + insight extraction)
- **Monthly (M1):** from 4-5 weeklies → monthly.md (index compression + cross-week patterns)
- **Yearly (Y1):** from 12 monthlies → yearly.md (annual narrative + theme evolution)

- **L4 Evolution (W2):** read weekly + SOUL/AGENTS/MEMORY → identity layer update

Parallel design: W1/M1/Y1 are three independent Protocol Builders. The weekly protocol automatically appends M1/Y1 when conditions align. Users can also trigger each protocol individually.

Deduplication matrix: six trigger paths with mutual exclusion—last-eod.json 30-minute window intercepts all. Atomic writes (tmp+rename) prevent concurrent corruption.

2.6 Active Triggering (V3.8.8-beta2)

Triangular factor: three factors vote on whether to execute:

$$\text{score} = 0.3 \times \text{keyword} + 0.4 \times \text{llm_intent} + 0.3 \times \text{state_relevance}$$

Factor	Role	Computation
keyword	deterministic anchor	hook-level regex matching, 0/1
llm_intent	semantic coverage	LLM tool-call behavior (called = 1)
state_relevance	calibration layer	sleepiness score / time since last EOD / file existence

"The dead handle the dead; the living handle the living"—keyword is a dead gate (zero false positives), llm_intent is a living semantic judgment, state is a calibrator. No single factor can trigger a protocol alone.

Three-level active invocation: when the LLM wants to execute a protocol, it cascades through fallbacks—L1 execute_protocol tool → L2 output marker on failure → L3 reads workflow docs to self-determine.

3. Implementation

3.1 Modular Architecture

27 TypeScript source files (6,273 lines), zero production dependencies. MENTIER analyzed memory retrieval bottlenecks and compression distortion in long-running agents within the same ecosystem [8]; this system adds differentiated rhythm-drive and personality-layering on top of that foundation:

```

text
src/
├─ index.ts          # Plugin entry (hooks + tool
registration)
├─ hub.ts            # Scene routing (startup / heartbeat /
recovery / conversation)
├─ protocol.ts       # Light / Medium / Full protocol
builders
├─ hooks/
│   ├─ before-prompt-build.ts  # Main hook
│   └─ before-tool-call.ts     # Safety interception
├─ modules/
│   └─ heartbeat.ts            # Heartbeat + sleepiness drive +
conversation path
│   └─ intent-resolver.ts      # Triangular factor intent
resolution
│   ├─ hardcoded/index.ts      # Script execution + state update
│   └─ memory/
│       ├─ reader.ts           # L2/L3 memory reading
│       └─ consolidation.ts    # Three parallel modules
├─ tools/
│   └─ execute-protocol.ts      # execute_protocol tool
└─ utils/
    ├─ agent-config.ts          # Three-parameter deployment
    └─ sleepiness-config.ts     # Config file reader

```

Daemon: scripts/sleepiness-watchdog.cjs (standalone Node.js process, setInterval 5 min, PID file + plugin keepalive self-healing).

3.2 Externalized Configuration

Zero-code deployment. Moving to a new agent requires editing three lines in openclaw.plugin.json and restarting the Gateway:

```

json
{
  "agentConfig": {
    "agentName": "your-agent-name",
    "dreamAgentName": "your-agents-dream",
    "extraBasePath": "/path/to/dialogue-logs"
  }
}

```

All sleepiness parameters centralized in config/sleepiness.json—weights, thresholds, distribution mappings, check script paths. Three consumers (heartbeat.ts, hardcoded/index.ts, sleepiness-watchdog.cjs) read the same file. Parameter changes require no recompilation.

3.3 Production Validation

- **Runtime:** WSL2 (Ubuntu 22.04) + OpenClaw Gateway v2026.3.13
- **Dual instances:** Yuka (primary) + Kayoko (independent instance, separate state directory)
- **Models:**
 - Yuka: DeepSeek V4 Pro (primary) / GLM-5 / Qwen3.5+ Alibaba Cloud CodingPlan-Lite
 - Kayoko: GLM-5 / GLM-5.1
- **Duration:** March–June 2026 (3 months)
- **Iteration:** 3 independent audit reports (31 issues, all closed)

Important: All production validation was conducted on cloud models (API calls). Local deployment (e.g., qwen3.6-MOE-A3B, qwen3.6-27B-dense) is theoretically compatible by design but has not undergone production-level verification. Local model effectiveness—particularly personality consistency and protocol adherence—requires independent evaluation.

3.4 Open-Source Infrastructure

MIT licensed. Sandbox package: 198 files, including 6 example identity layer files, one-command deployment script (quickstart.sh), bilingual documentation (Chinese/English).

4. Discussion

4.1 Design for Cloud API-Driven Agents

Mem0 is a general-purpose Agent memory layer with performance data on the LoCoMo benchmark [9]. Soul Protocol addresses different needs: memory serves personality continuity rather than task continuity, with archiving triggered automatically by system state.

Design assumptions for cloud API models:

- **No assumption of perfect semantic understanding** — the triangular factor relies on both keyword anchors and LLM intent.

- **No assumption of permanent long-context access** — each layer functions independently.
- **Zero training cost** — personality is defined through text files, not fine-tuning.
- **Local models await validation** — this plugin has **only been validated on cloud models**.

4.2 Limitations

1. **Uncalibrated thresholds** — sleepiness weights and thresholds are heuristic, not statistically validated.
2. **Missing facts layer** — identity layer lacks a TOOLS/NOTES layer (see §2.4.1), causing routing errors.
3. **No quantitative evaluation** — personality consistency and memory accuracy have not been benchmarked.
4. **Single-platform dependency** — only validated on OpenClaw.
5. **Weak assistant capabilities** — excels at companionship and memory, unsuitable for precise information management.

4.3 Future Work (V3.9)

- Multi-agent relationship design
- Long-form narrative memory injection
- TOOLS.md facts layer (identity layer sixth file)
- Dreaming protocol deepening

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